PREDICTION IN THE SIMPLE LINEAR REGRESSION MODEL

Learning Outcomes

- describe the use of analysis of variance (ANOVA) in regression analysis, interpret ANOVA results, and calculate and interpret the standard error of estimate in a simple linear regression
- calculate and interpret the predicted value for the dependent variable, and a prediction interval for it, given an estimated linear regression model and a value for the independent variable

ANOVA and Standard Error of Estimate in Simple Linear Regression

We often represent the sums of squares from a regression model in an **analysis of variance (ANOVA)** table, as shown in Exhibit 32, which presents the sums of squares, the degrees of freedom, the mean squares, and the *F*-statistic. Notice that the variance of the dependent variable is the ratio of the sum of squares total to $n - 1$.

Exhibit 32: Analysis of Variance Table for Simple Linear Regression

Source	Sum of Squares	Degrees of Freedom	Mean Square	*F*-Statistic
Regression	$\text{SSR} = \sum_{i=1}^{n}\left(\hat{Y}_i - \bar{Y}\right)^2$	1	$\text{MSR} = \frac{\sum_{i=1}^{n}\left(\hat{Y}_i - \bar{Y}\right)^2}{1}$	$F = \frac{\text{MSR}}{\text{MSE}} = \frac{\frac{\sum_{i=1}^{n}\left(\hat{Y}_i - \bar{Y}\right.}{1}}{\frac{\sum_{i=1}^{n}\left(Y_i - \hat{Y}_i\right.}{n-2}}$
Error	$\text{SSE} = \sum_{i=1}^{n}\left(Y_i - \hat{Y}_i\right)^2$	$n - 2$	$\text{MSE} = \frac{\sum_{i=1}^{n}\left(Y_i - \hat{Y}_i\right)^2}{n-2}$	
Total	$\text{SST} = \sum_{i=1}^{n}\left(Y_i - \bar{Y}\right)^2$	$n - 1$		

From the ANOVA table, we can also calculate the **standard error of the estimate** (s_e), which is also known as the standard error of the regression or the root mean square error. The s_e is a measure of the distance between the observed values of the dependent variable and those predicted from the estimated regression—the smaller the s_e, the better the fit of the model. The s_e, along with the coefficient of determination and the *F*-statistic, is a measure of the goodness of the fit of the estimated regression line. Unlike the coefficient of determination and the *F*-statistic, which are relative measures of fit, the standard error of the estimate is an absolute measure of the distance of the observed dependent variable from the regression line. Thus, the s_e is an important statistic used to evaluate a regression model and is used to calculate prediction intervals and to perform tests on the coefficients. The calculation of s_e is straightforward once we have the ANOVA table because it is the square root of the MSE:

$$\text{Standard error of the estimate } (s_e) = \sqrt{\text{MSE}} = \sqrt{\frac{\sum_{i=1}^{n}\left(Y_i - \hat{Y}_i\right)^2}{n-2}}. \quad 22$$

We show the ANOVA table for our ROA regression example in Exhibit 32, using the information from Exhibit 33. For a 5 percent level of significance, the critical *F*-value for the test of whether the model is a good fit (i.e., whether the slope coefficient is different from zero) is 7.71. We can get this critical value in the following ways:

- *Excel* [F.INV(0.95,1,4)]
- *R* [qf(.95,1,4)]
- *Python* [from scipy.stats import f *and* f.ppf(.95,1,4)]

With a calculated *F*-statistic of 16.0104 and a critical *F*-value of 7.71, we reject the null hypothesis and conclude that the slope of our simple linear regression model for ROA is different from zero.

Exhibit 33: ANOVA Table for ROA Regression Model

Source	Sum of Squares	Degrees of Freedom	Mean Square	*F*-Statistic
Regression	191.625	1	191.625	16.0104
Error	47.875	4	11.96875	
Total	239.50	5		

The calculations to derive the ANOVA table and ultimately to test the goodness of fit of the regression model can be time consuming, especially for samples with many observations. However, statistical packages, such

Financial analysts often want to use regression results to make predictions about a dependent variable. For example, we might ask, "How fast will the sales of XYZ Corporation grow this year if real GDP grows by 4 percent?" But we are not merely interested in making these forecasts; we also want to know how certain we can be about the forecasts' results. A forecasted value of the dependent variable, $\widehat{Y}_f$, is determined using the estimated intercept and slope, as well as the expected or forecasted independent variable, X_f:

$$\widehat{Y}_f = \widehat{b}_0 + \widehat{b}_1 X_f. \qquad 23$$

In our ROA regression model, if we forecast a company's CAPEX to be 6 percent, the forecasted ROA based on our estimated equation is 12.375 percent:

$$\widehat{Y}_f = 4.875 + (1.25 \times 6) = 12.375.$$

However, we need to consider that the estimated regression line does not describe the relation between the dependent and independent variables perfectly; it is an average of the relation between the two variables. This is evident because the residuals are not all zero.

Therefore, an interval estimate of the forecast is needed to reflect this uncertainty. The estimated variance of the prediction error, s_f^2, of *Y*, given *X*, is

$$s_f^2 = s_e^2\left[1 + \frac{1}{n} + \frac{(X_f - \overline{X})^2}{(n-1)s_X^2}\right] = s_e^2\left[1 + \frac{1}{n} + \frac{(X_f - \overline{X})^2}{\sum_{i=1}^{n}(X_i - \overline{X})^2}\right],$$

and the **standard error of the forecast** is

$$s_f = s_e\sqrt{1 + \frac{1}{n} + \frac{(X_f - \overline{X})^2}{\sum_{i=1}^{n}(X_i - \overline{X})^2}}. \qquad 24$$

The standard error of the forecast depends on the following:

- the standard error of the estimate, s_e;
- the number of observations, *n*;
- the forecasted value of the independent variable, X_f, used to predict the dependent variable and its deviation from the estimated mean, $\overline{X}$; and
- the variation of the independent variable.

We can see the following from the equation for the standard error of the forecast:

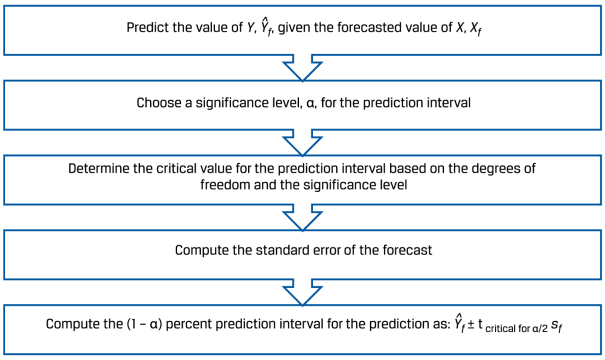
1. The better the fit of the regression model, the smaller the standard error of the estimate (s_e) and, therefore, the smaller standard error of the forecast.
2. The larger the sample size (*n*) in the regression estimation, the smaller the standard error of the forecast.
3. The closer the forecasted independent variable (X_f) is to the mean of the independent variable ($\overline{X}$) used in the regression estimation, the smaller the standard error of the forecast.

Once we have this estimate of the standard error of the forecast, determining a prediction interval around the predicted value of the dependent variable ($\widehat{Y}_f$) is very similar to estimating a confidence interval around an estimated parameter. The prediction interval is

$$\widehat{Y}_f \pm t_{critical\ for\ \alpha/2} s_f.$$

We outline the steps for developing the prediction interval in Exhibit 34.

Exhibit 34: Creating a Prediction Interval around the Predicted Dependent Variable



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Assuming a 5 percent significance level (α), two sided, with $n - 2$ degrees of freedom (so, df = 4), the critical values for the prediction interval are ±2.776.

The standard error of the forecast is

$$s_f = 3.459588\sqrt{1 + \frac{1}{6} + \frac{(6 - 6.1)^2}{122.640}} = 3.459588\sqrt{1.166748} = 3.736912.$$

The 95 percent prediction interval then becomes

$$12.375 \pm 2.776\ (3.736912)$$

$$12.375 \pm 10.3737$$

$$\left\{2.0013 < \widehat{Y}_f < 22.7487\right\}$$

For our ROA regression example, we can see how the standard error of the forecast (s_f) changes as our forecasted value of the independent variable gets farther from the mean of the independent variable $(X_f - \overline{X})$ in Exhibit 35. The mean of CAPEX is 6.1 percent, and the band that represents one standard error of the forecast, above and below the forecast, is minimized at that point and increases as the independent variable gets farther from $\overline{X}$.

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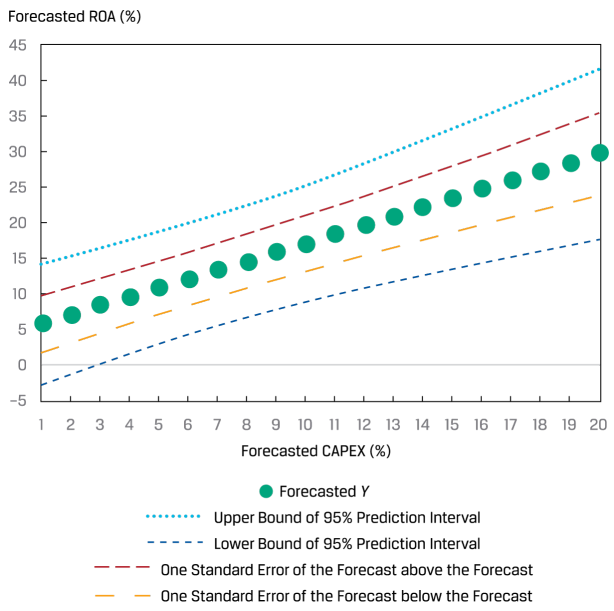
Exhibit 35: ROA Forecasts and Standard Error of the Forecast



Question Set

Suppose you run a cross-sectional regression for 100 companies, where the dependent variable is the annual return on stock and the independent variable is the lagged percentage of institutional ownership (INST). The results of this simple linear regression estimation are shown in Exhibit 36. Evaluate the model by answering questions 1–4.

Exhibit 36: ANOVA Table for Annual Stock Return Regressed on Institutional Ownership

Source	Sum of Squares	Degrees of Freedom	Mean Square
Regression	576.1485	1	576.1485
Error	1,873.5615	98	19.1180
Total	2,449.7100		

1. What is the coefficient of determination for this regression model?

Hide Solution

Solution:

The coefficient of determination is sum of squares regression/sum of squares total: 576.148 ÷ 2,449.71 = 0.2352, or 23.52 percent.

The standard error of the estimate is the square root of the mean square error: $\sqrt{19.1180} = 4.3724$.

3. At a 5 percent level of significance, do we reject the null hypothesis of the slope coefficient equal to zero if the critical *F*-value is 3.938?

Hide Solution

Solution:

Using a six-step process for testing hypotheses, we get the following:

Step 1	State the hypotheses.	H_0: $b_1 = 0$ versus H_a: $b_1 \neq 0$
Step 2	Identify the appropriate test statistic.	$F = \frac{\text{MSR}}{\text{MSE}}$ with 1 and 98 degrees of freedom.
Step 3	Specify the level of significance.	α = 5% (one tail, right side).
Step 4	State the decision rule.	Critical *F*-value = 3.938. Reject the null hypothesis if the calculated *F*-statistic is greater than 3.938.
Step 5	Calculate the test statistic.	$F = \frac{576.1485}{19.1180} = 30.1364$
Step 6	Make a decision.	Reject the null hypothesis because the calculated *F*-statistic is greater than the critical *F*-value. Evidence is sufficient to indicate that the slope coefficient is different from 0.0.

4. Based on your answers to the preceding questions, evaluate this simple linear regression model.

Hide Solution

Solution:

The coefficient of determination indicates that variation in the independent variable explains 23.52 percent of the variation in the dependent variable. Also, the *F*-statistic test confirms that the model's slope coefficient is different from 0 at the 5 percent level of significance. In sum, the model seems to fit the data reasonably well.

The following applies to questions 5-11.

Suppose we want to forecast a company's net profit margin (NPM) based on its research and development expenditures scaled by revenues (RDR), using the model estimated and the details provided in Exhibit 8. The regression model was estimated using data on eight companies as

$$\widehat{Y}_f = 16.5 - 1.3X_f,$$

with a standard error of the estimate (s_e) of 1.8618987 and variance of RDR, $\frac{\sum_{i=1}^{n}(X_i - \overline{X})^2}{(n-1)}$, of 4.285714, as given.

5. What is the predicted value of NPM if the forecasted value of RDR is 5?

Hide Solution

Solution:

The predicted value of NPM is 10: 16.5 − (1.3 × 5) = 10.

6. What is the standard error of the forecast (s_f) if the forecasted value of RDR is 5?

Hide Solution

Solution:

To derive the standard error of the forecast (s_f), we first have to calculate the variation of RDR. Then, we have all the pieces to calculate s_f:

$$\sum_{i=1}^{n}(X_i - \overline{X})^2 = 4.285714 \times 7 = 30.$$

$$s_f = 1.8618987\sqrt{1 + \frac{1}{8} + \frac{(5-7.5)^2}{30}} = 2.1499.$$

7. What is the 95 percent prediction interval for the predicted value of NPM using critical *t*-values (df = 6) of ±2.447?

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$$\{10 \pm 2.447(2.1499)\},$$

$$\left\{4.7392 < \widehat{Y}_f < 15.2608\right\}.$$

8. What is the predicted value of NPM if the forecasted value of RDR is 15?

Hide Solution

Solution:

The predicted value of NPM is −3: 16.5 − (1.3 × 15) = −3.

9. Referring to exhibit 9, what is the standard error of the forecast if the forecasted value of RDR is 15?

Hide Solution

Solution:

To derive the standard error of the forecast, we first must calculate the variation of RDR. Then, we can calculate s_f:

$$\sum_{i=1}^{n} \left(X_i - \overline{X}\right)^2 = 4.285714 \times 7 = 30.$$

$$s_f = 1.8618987\sqrt{1 + \frac{1}{8} + \frac{(15 - 7.5)^2}{30}} = 3.2249.$$

10. What is the 95 percent prediction interval for the predicted value of NPM using critical *t*-values (df = 6) of ±2.447?

Hide Solution

Solution:

The 95 percent prediction interval for the predicted value of NPM is

$$\{-3 \pm 2.447(3.2249)\},$$

$$\left\{-10.8913 < \widehat{Y}_f < 4.8913\right\}.$$

11. Calculate the sample variance of the dependent variable using information in the table.

Hide Solution

Solution:

The sample variance of the dependent variable is the sum of squares total divided by its degrees of freedom (n − 1 = 5 − 1 = 4, as given). Thus, the sample variance of the dependent variable is 95.2 ÷ 4 = 23.8.

The following applies to questions 12-15.

You are examining the results of a regression estimation that attempts to explain the unit sales growth of a business you are researching. The analysis of variance output for the regression is given in the Exhibit 37. The regression was based on five observations (n = 5).

Exhibit 37: ANOVA Output

Source	df	Sum of Squares	Mean Square	*F*	*p*-Value
Regression	1	88.0	88.0	36.667	0.00904
Residual	3	7.2	2.4		
Total	4	95.2			

12. Calculate the coefficient of determination for this estimated model.

Hide Solution

Solution:

The coefficient of determination = 88.0 ÷ 95.2 = 0.92437.

13. What hypothesis does the *F*-statistic test?

Hide Solution

Solution:

The *F*-statistic tests whether all the slope coefficients in a linear regression are equal to zero.

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The calculated value of the *F*-statistic is 36.667, as shown in Exhibit 32. The corresponding *p*-value is less than 0.05, so you reject the null hypothesis of a slope equal to zero.

15. Calculate the standard error of the estimate.

Hide Solution

Solution:

The standard error of the estimate is the square root of the mean square error: $s_e = \sqrt{2.4}$ = 1.54919.

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MO

Q11, I'm so confused here, the sum of squares total comes to 71.5 (SSR 50.7 + SSE 20.8); where did they get 95.2? Moreover there are 8 companies in that table, not 5, so why do we have n - 1 = 5 - 1, instead of 8 - 1? I thought, maybe it's the ROA vs CAPEX example, but there are 6 companies in that table, not 5...

Created 5 months ago by Maxim Ovchinnikov — 3 replies | Last Activity: 2 months ago

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MO

The answer in Q6 is incorrect, sf = 1.8618987 * 1.333333 = 2.4825232, but they have 2.1499?

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IZ

In question 6 I miss the mean number in equation. I see that it is 7,5 from solution, but I have no idea of how to calculate it. Is this a mistake, or am I missing something?

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